Zack Abdirashid
2/25/26
MATH 402

<u>**5**</u>

1. Consider the following permutations given in array notation:

$$\alpha = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 2 & 3 & 4 & 5 & 1 & 7 & 8 & 6 \end{bmatrix} \qquad \beta = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 3 & 8 & 7 & 6 & 5 & 2 & 4 \end{bmatrix}$$

(a) Write α, β, and $\alpha\beta$ as products of disjoint cycles.

$$\boxed{\alpha = (12345)(678)}$$
$$\boxed{\beta = (23847)(56)}$$

$*\alpha\beta$ uses function composition

$$\Rightarrow \boxed{\alpha\beta = (12485736)}$$

(b) Compute each of the inverses α^{-1}, β^{-1}, and $(\alpha\beta)^{-1}$.

* Taking the inverse of a permutation $\alpha \Rightarrow$ reverse the original permutation

$$\boxed{\alpha^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 1 & 2 & 3 & 4 & 8 & 6 & 7 \end{bmatrix}}$$

$$\boxed{\beta^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 1 & 7 & 2 & 8 & 6 & 5 & 4 & 3 \end{bmatrix}}$$

$$\boxed{(\alpha\beta)^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 6 & 1 & 7 & 2 & 8 & 3 & 5 & 4 \end{bmatrix}}$$

(c) Determine the order of each of α, β, and $\alpha\beta$.

* Order of a permutation $\Rightarrow \text{lcm}(\text{len}(c_1), \text{len}(c_2), \ldots, \text{len}(c_n))$

$$\begin{aligned} |\alpha| &= \text{lcm}(\text{len}(12345), \text{len}(678)) \\ &= \text{lcm}(5,3) = \boxed{15} \end{aligned}$$

$$\begin{aligned} |\beta| &= \text{lcm}(\text{len}(23847), \text{len}(56)) \\ &= \text{lcm}(5,\ 2) = \boxed{10} \end{aligned}$$

$$\begin{aligned} |\alpha\beta| &= \text{lcm}(\text{len}(12485736)) \\ &= \text{lcm}(8) = \boxed{8} \end{aligned}$$

(d) Determine whether each of α, β, and $\alpha\beta$ is even or odd.

* Is the # of transpositions even or odd?

$$\alpha = \underbrace{(15)(14)(13)(12)(68)(67)}_{\text{6 transpositions}}$$
$$\Rightarrow \boxed{\alpha \text{ is even}}$$

$$\beta = \underbrace{(27)(24)(28)(23)(56)}_{\text{5 transpositions}}$$
$$\Rightarrow \boxed{\beta \text{ is odd}}$$

$$\alpha\beta = \underbrace{(16)(13)(17)(15)(18)(14)(12)}_{\text{7 transpositions}}$$
$$\Rightarrow \boxed{\alpha\beta \text{ is odd}}$$

2. (a) Prove that the alternating group A_n is a subgroup of the symmetric group S_n.

Proof. Let even permutations $p_i, p_j \in A_n$. By the One-Step Subgroup Test, we need to show that $p_i(p_j)^{-1} \in A_n$ for all $p_i, p_j \in A_n$ to prove that $A_n \leq S_n$. Taking the inverse of an even permutation reverses the mapping of the original, but the length is unchanged. So,

$$p_i(p_j)^{-1} = p_i p_k \text{ where } p_k \in A_n.$$

Since p_i and p_k are even permutations, both can be expressed as an even # of transpositions. So, the composition of $p_i p_k$ results in the sum of their total transpositions, which is also even. This makes $p_i p_k$ an even permutation. So,

$$p_i p_k = p_i(p_j)^{-1} \in A_n.$$

By the One-Step Subgroup Test, $A_n \leq S_n$. □

(b) Let α, $\beta \in S_n$. Show that $\beta\alpha\beta^{-1}$ and α are either both even or both odd.

* For a permutation to be even/odd, it can be written as an even/odd # of transpositions.

<u>**Case** 1: α is even</u>

If β is even,

$$\begin{aligned}\beta\alpha\beta^{-1} &= (\text{even \# of 2-cycles}) + (\text{even \# of 2-cycles}) + (\text{even \# of 2-cycles}) \\ &= \text{even \# of transpositions}\end{aligned}$$

If β is odd,

$$\begin{aligned}\beta\alpha\beta^{-1} &= (\text{odd \# of 2-cycles}) + (\text{even \# of 2-cycles}) + (\text{odd \# of 2-cycles})\\ &= (\text{odd \# of 2-cycles}) + (\text{odd \# of 2-cycles})\\ &= \text{even \# of transpositions}\end{aligned}$$

$\Rightarrow \ \beta\alpha\beta^{-1}$is even.

Case 2: α is odd

If β is even,

$$\begin{aligned}\beta\alpha\beta^{-1} &= (\text{even \# of 2-cycles}) + (\text{odd \# of 2-cycles}) + (\text{even \# of 2-cycles})\\ &= (\text{odd \# of 2-cycles}) + (\text{even \# of 2-cycles})\\ &= \text{odd \# of transpositions}\end{aligned}$$

If β is odd,

$$\begin{aligned}\beta\alpha\beta^{-1} &= (\text{odd \# of 2-cycles}) + (\text{odd \# of 2-cycles}) + (\text{odd \# of 2-cycles})\\ &= \text{odd \# of transpositions}\end{aligned}$$

$\Rightarrow \ \beta\alpha\beta^{-1}$is odd

3. (a) Consider the sets

$$G_1 = \left\{a + b\sqrt{2} \;\middle|\; a, b \in \mathbb{Q}\right\} \text{ and } G_2 = \left\{\begin{bmatrix} a & 2b \\ b & a \end{bmatrix} \;\middle|\; a, b \in \mathbb{Q}\right\}$$

Both G_1 and G_2 are groups under the operation of addition. Prove that G_1 and G_2 are isomorphic.

* "isomorphism" $\Rightarrow$ bijective and operation-preserving Since both G_1 and G_2 make use of a and b, define $\varphi : G_1 \to G_2$ to be

$$\varphi(a + b\sqrt{2}) = \begin{bmatrix} a & 2b \\ b & a \end{bmatrix}.$$

One-to-One: Suppose we have some $a_1, b_1 \in \mathbb{Q}$ and $a_2, b_2 \in \mathbb{Q}$ s.t

$$\varphi(a_1 + b_1\sqrt{2}) = \varphi(a_2 + b_2\sqrt{2})$$

$$\Rightarrow \begin{bmatrix} a_1 & 2b_1 \\ b_1 & a_1 \end{bmatrix} = \begin{bmatrix} a_2 & 2b_2 \\ b_2 & a_2 \end{bmatrix}.$$

However, for matrices to be equal, they must have the same dimensions and the same elements for each row and column. So,

$$a_1 = a_2$$

$$\begin{aligned} 2b_1 &= 2b_2 \\ \Rightarrow \ b_1 &= b_2 \end{aligned}$$

So, $a_1 + b_1\sqrt{2} = a_2 + b_2\sqrt{2}$, making φ 1-to-1.

Onto: Let $M = \begin{bmatrix} a_1 & 2b_1 \\ b_1 & a_1 \end{bmatrix} \in G_2$ where $a_1, b_1 \in \mathbb{Q}$. Since $a_1, b_1 \in \mathbb{Q}$, there must be some $g \in G_1$ s.t $g = a_1 + b_1\sqrt{2}$. But, that implies

$$\varphi(g) = M.$$

So, any matrix M in G_2 has a corresponding element in G_1 that maps to it, making φ onto.

Operation Preserving: Let $a_1, a_2, b_1, b_2 \in \mathbb{Q}$. Consider

$$\begin{aligned} \varphi(a_1 + b_1\sqrt{2}) + \varphi(a_2 + b_2\sqrt{2}) &= \begin{bmatrix} a_1 & 2b_1 \\ b_1 & a_1 \end{bmatrix} + \begin{bmatrix} a_2 & 2b_2 \\ b_2 & a_2 \end{bmatrix} \\ &= \begin{bmatrix} a_1 + a_2 & 2b_1 + 2b_2 \\ b_1 + b_2 & a_1 + a_2 \end{bmatrix} \\ &= \begin{bmatrix} a_1 + a_2 & 2(b_1 + b_2) \\ b_1 + b_2 & a_1 + a_2 \end{bmatrix} \\ &= \varphi(a_1 + b_1\sqrt{2} + a_2 + b_2\sqrt{2}). \end{aligned}$$

So, φ is operation preserving. $\therefore G_1 \cong G_2$.

(b) Prove that the group $G_1 = (\mathbb{Z}, +)$ and $G_2 = (\mathbb{Q}, +)$ are **not** isomorphic.

Proof. By contradiction, suppose $\exists$ an isomorphism $\varphi : G_1 \to G_2$. That implies G_1 <u>and</u> G_2 share the same properties of groups such as being cyclic. We know $G_1 = \langle 1 \rangle$. However, there's no rational number $\frac{a}{b}$ that generates every rational number under addition because b is fixed as the denominator which makes it impossible to reach any other rational number whose denominator isn't b. So, $G_1 \ncong G_2$. $\square$

4. (a) Prove that group isomorphism is an equivalence relation.

Proof. To show that group isomorphism is an equivalence relation, we must show that it satisfies the properties of one.

Reflexivity: Let G be a group. We want to show that $\exists$ an isomorphism between any G and itself. We can define

$$\varphi_a = axa^{-1} \text{ where } a \in G$$

as the inner automorphism of G induced by a. This allows us to say that the function

$$\varphi_a : G \to G$$

is an isomorphism between G and itself.

Symmetry: Let G_1 and G_2 be groups and $\varphi : G_1 \to G_2$ be an isomorphism. We want to show $\exists$ some isomorphism mapping G_2 back to G_1. By Theorem 6.3, the function

$$\varphi^{-1} : G_2 \to G_1$$

is also an isomorphism which maps G_2 back to G_1.

Transitivity: Let G_1, G_2 and G_3 be groups and $\varphi : G_1 \to G_2$ and $\pi : G_2 \to G_3$ be isomorphisms. We want to show that $\exists$ an isomorphism between G_1 to G_3. Consider the function

$$\pi \circ \varphi : G_1 \to G_3.$$

Let $a, b \in G_1$ s.t $\pi \circ \varphi(a) = \pi \circ \varphi(b)$. Since φ and π are isomorphisms, they're 1-to-1. So,

$$\begin{aligned} \pi(\varphi(a)) &= \pi(\varphi(b)) \\ \Rightarrow \pi(a) &= \pi(b) \\ \Rightarrow a &= b. \end{aligned}$$

So, $\pi \circ \varphi$ is 1-to-1. Let $d \in G_2$. Since φ is an isomorphism, we know there exists some $c \in G_1$ s.t

$$\varphi(c) = d.$$

Since π is an isomorphism, there's some $e \in G_3$ s.t

$$\begin{aligned} \pi(\varphi(c)) &= \pi(d) \\ &= e. \end{aligned}$$

So, $\pi \circ \varphi$ is onto. To show this composition is operation preserving, we need to show that $\pi \circ \varphi(ab) = [\pi \circ \varphi(a)][\pi \circ \varphi(b)]$. Since φ and π are isomorphisms,

$$\begin{aligned} \pi \circ \varphi(ab) &= \pi(\varphi(ab)) \\ &= \pi(\varphi(a)\varphi(b)) \\ &= [\pi(\varphi(a)][\pi(\varphi(b)] \\ &= [\pi \circ \varphi(a)][\pi \circ \varphi(b)]. \end{aligned}$$

Hence, $\pi \circ \varphi$ is an isomorphism and $G_1 \cong G_3$. So, group isomorphism is an equivalence relation. $\square$

(b) If a group G is isomorphic to a group H, prove that $\mathrm{Aut}(G)$ is isomorphic to $\mathrm{Aut}(H)$.

Zack Abdirashid
2/25/26
MATH 402

Proof. Suppose $\exists$ an isomorphism $\varphi : G \to H$. Consider the mapping

$$\Phi : \mathrm{Aut}(G) \to \mathrm{Aut}(H)$$

given by $\Phi(x) = \varphi\alpha\varphi^{-1}$. To show this mapping is an isomorphism, we must show that it is bijective and operation preserving.

<u>One-to-One</u>: Suppose $\alpha_1, \alpha_2 \in \mathrm{Aut}(G)$ such that

$$\begin{gathered}\Phi(\alpha_1) = \Phi(\alpha_2)\\ \Rightarrow \varphi\alpha_1\varphi^{-1} = \varphi\alpha_2\varphi^{-1}.\end{gathered}$$

We want to show that $\alpha_1 = \alpha_2$. Since φ is an isomorphism, $\varphi^{-1} : H \to G$ is also an isomorphism. So,

$$\begin{aligned}&\varphi^{-1}(\varphi\alpha_1\varphi^{-1})\varphi = \varphi^{-1}(\varphi\alpha_2\varphi^{-1})\varphi\\ \Rightarrow\ &\varphi^{-1}\varphi(\alpha_1)\varphi^{-1}\varphi = \varphi^{-1}\varphi(\alpha_2)\varphi^{-1}\varphi\\ &\Rightarrow \alpha_1 = \alpha_2.\end{aligned}$$

So, Φ is 1-to-1.

<u>Onto</u>: We want to show that for some $a \in \mathrm{Aut}(H)$ that

$$\begin{aligned}\Phi(\alpha) &= a\\ \Rightarrow \varphi\alpha\varphi^{-1} &= a\\ \Rightarrow \varphi^{-1}a\ \varphi &= \alpha \text{ where } \varphi^{-1}a\ \varphi \in \mathrm{Aut}(G).\end{aligned}$$

So,

$$\begin{aligned}\Phi(\alpha) &= \varphi(\varphi^{-1}a\ \varphi)\varphi^{-1}\\ &= \varphi\varphi^{-1}(a)\varphi\varphi^{-1}\\ &= a.\end{aligned}$$

Hence, Φ is onto.

<u>Operation Preserving</u>: To show that $\varphi(\alpha_1\alpha_2) = \varphi(\alpha_1)\varphi(\alpha_2)$, consider

$$\begin{aligned}\Phi(\alpha_1)\Phi(\alpha_2) &= (\varphi\alpha_1\varphi^{-1})(\varphi\alpha_2\varphi^{-1})\\ &= \varphi\alpha_1(\varphi^{-1}\varphi)\alpha_2\varphi^{-1}\\ &= \varphi\alpha_1\alpha_2\varphi^{-1}\\ &= \Phi(\alpha_1\alpha_2).\end{aligned}$$

So, Φ is operation preserving. $\therefore$ Φ is an isomorphism and $\mathrm{Aut}(G) \cong \mathrm{Aut}(H)$. $\square$

(c) Give an example of two groups G and H such that $\text{Aut}(G)$ is isomorphic to $\text{Aut}(H)$, but G is **not** isomorphic with H.

Consider $\mathbb{Z}_4$ and $\mathbb{Z}_6$. By Theorem 6.5, we know that

$$\text{Aut}(\mathbb{Z}_4) \cong U(4) = \{1, 3\}$$
$$\text{Aut}(\mathbb{Z}_6) \cong U(6) = \{1, 5\}.$$

Since both sets of automorphisms are isomorphic to cyclic groups of order 2,

$$\text{Aut}(\mathbb{Z}_4) \cong \text{Aut}(\mathbb{Z}_6).$$

But, $|\mathbb{Z}_4| = 4 \neq |\mathbb{Z}_6| = 6$, making a 1-to-1 mapping between these groups impossible. Hence, $\mathbb{Z}_4 \ncong \mathbb{Z}_6$.